

STC PORTAL

College Placement Intelligence Platform, KIET Group of Institutions

Problem Identification and Feasibility Analysis

1. Introduction

The STC Portal is a web-based placement intelligence platform developed for KIET Group of Institutions to centralize, verify, and present the college's placement records to its own students in a secure, self-service format. The platform replaces informal, unverifiable channels — WhatsApp forwards, shared spreadsheets, and word-of-mouth compensation figures — with a single authenticated system of record, backed by a Supabase PostgreSQL database and a React front end capable of serving both light and dark interface themes across desktop and mobile devices.

This document identifies the core problem the STC Portal was built to solve, defines the objectives and scope of the software project, and presents a feasibility analysis covering the technical and operational dimensions of building and sustaining the platform within the Training and Placement Cell (TPC) of the college.

2. Problem Identification

2.1 Background

Prior to the STC Portal, placement information at the college was distributed across disconnected, informal channels maintained independently by student volunteers, class representatives, and the placement cell. Historical CTC (Cost to Company) figures, company visit records, and interview experiences existed only as scattered PDFs, chat exports, and personal notes, with no consolidated archive spanning multiple academic years.

2.2 Problem Statement

Students at KIET Group of Institutions lack a single, verified, and access-controlled source of placement data for their own college — resulting in unreliable CTC figures, no structured history of past recruitment cycles, no standardized way to practice for upcoming drives, and a persistent risk of one college's private placement data being exposed to students from other institutions when generic, shared tools are used instead.

2.3 Contributing Issues

- Fragmented records: placement reports were stored in personal drives and group chats, with no searchable archive organized by year or company.
- No verification mechanism: any student or contributor could report a CTC figure with no review step, making the data unreliable for juniors preparing for interviews.

- No structured preparation tools: historical OA (Online Assessment) questions and interview experiences were not linked to a practice workflow, so students could not rehearse against real, past company patterns.
- Cross-institution data exposure: generic, open-access placement trackers used by multiple colleges created a risk of one institution's placement figures being visible to students outside that institution.
- Administrative overhead: placement coordinators had no dashboard to add, verify, or retract company and offer data, forcing manual, ad hoc updates to shared documents.
- Scalability concerns: on the day a shortlist is announced, informal channels (chat groups, shared documents) cannot handle simultaneous access from the entire eligible batch without becoming unusable.

3. Objectives

The STC Portal is guided by the following primary objectives:

1. Centralize placement data. Consolidate every verified company visit, offer, and CTC figure into one searchable, year-wise archive, replacing scattered informal records.
2. Verify before publishing. Route every submitted placement report through an admin approval workflow (Approve / Reject / Edit) so that only verified figures are visible to students.
3. Restrict access by institution. Authenticate users through Google Sign-In restricted to each college's own email domain, ensuring a student can view only their own institution's placement data.
4. Tier administrative access. Allow any email address to be granted admin privileges, but only after explicit whitelisting by a Super Admin, preventing unauthorized data modification.
5. Support interview preparation. Provide a Mock Practice module and an OA (Online Assessment) question bank drawn from verified historical data, turning past placement records into active preparation material.
6. Deliver a consistent, responsive interface. Offer both dark and light themes with a single, mobile-responsive layout so the experience is identical in quality on a laptop or a phone.
7. Remain performant under load. Sustain simultaneous access from an entire eligible student batch during high-traffic events such as shortlist announcements, through a multi-layer caching architecture.
8. Enable data-informed operation. Integrate Google Analytics so the Training and Placement Cell can observe real usage patterns and prioritize future improvements with evidence rather than assumption.

4. Scope

4.1 In Scope

- Student-facing dashboard displaying total reports, highest package, companies visited, reports-per-year trends, CTC distribution by bracket, and tier-wise (Elite / Major / Core) company breakdown.

- Company Reports module listing individual placement records with company name, role, CTC, year, and approval status.
- Mock Practice module and OA Questions bank for structured interview and assessment preparation.
- Google Sign-In authentication restricted to a college's registered email domain for student accounts.
- Super Admin—controlled whitelisting mechanism permitting any email address to be elevated to an Admin role.
- Admin Panel supporting company management, report management (approve / reject / edit / delete), OA question management, and admin whitelisting, with an Analytics tab for Super Admins.
- Dark and light theme support with a fully responsive layout for mobile and desktop browsers.
- A four-layer caching architecture (client, edge/CDN, application, and database layers) to sustain performance during high-concurrency events.
- Google Analytics integration for usage tracking and feature-adoption insight.
- Migration of legacy placement records from MySQL into the Supabase PostgreSQL backend.

4.2 Out of Scope

- Direct integration with external recruiter or ATS (Applicant Tracking System) platforms.
- Automated resume screening, shortlisting, or scoring of candidates.
- Real-time chat or messaging between students and recruiters.
- Cross-institution data sharing or a multi-tenant view spanning more than one college in the current release.
- Payment processing or any monetary transaction handling.
- Native mobile applications (the platform is delivered as a responsive web application only).

5. Feasibility Analysis

Feasibility analysis was conducted along two dimensions relevant to a college-operated software project: technical feasibility, examining whether the proposed architecture, tools, and skills can deliver the system as scoped, and operational feasibility, examining whether the platform can be adopted, staffed, and sustained within the day-to-day working of the Training and Placement Cell.

5.1 Technical Feasibility

Technical feasibility assesses whether the chosen technology stack, architecture, and available development skills are sufficient to build and operate the system as scoped.

Component	Technology / Approach	Feasibility Justification
Front end	React with Vite build tooling	Supports component reuse across dashboard, admin panel, and mock-practice views; Vite provides fast builds suited to a small

Component	Technology / Approach	Feasibility Justification
		development team; enables the dual dark/light theme system and responsive layout from a single codebase.
Backend / Database	Supabase (managed PostgreSQL)	Provides authentication, row-level security, and a relational database without requiring the college to provision or maintain its own database server; legacy MySQL data was migrated in directly.
Authentication	Google Sign-In with domain restriction	Google's OAuth infrastructure is mature and free to integrate; restricting accepted sign-in domains to a college's own email domain is a standard, well-documented configuration rather than custom-built security logic.
Caching layer	Four-layer cache cascade (client, edge/CDN, application, database)	Each layer uses established, well-supported caching patterns; the cascade is designed specifically to absorb simultaneous access spikes on shortlist-announcement days without requiring paid infrastructure scaling.
Analytics	Google Analytics	Free-tier integration with a single tracking script; no custom analytics pipeline needs to be built or maintained.
Hosting	Standard static/edge hosting for a Vite-built React app	React/Vite output is a static bundle compatible with widely available, low-cost edge hosting providers.

Risk assessment: the highest technical risk lies in sustaining performance during concentrated traffic spikes (e.g., the moment a shortlist is published). This is mitigated directly by the four-layer cache cascade, which serves the large majority of dashboard and report reads from cache rather than the database, reducing load on the Supabase backend during peak windows. Domain-restricted authentication and the admin-whitelisting mechanism are both configuration-level controls on top of Google's OAuth service rather than custom-built authentication logic, which keeps the associated security risk and implementation effort low. Overall, the project is judged technically feasible: every required capability maps to a mature, well-documented technology or pattern, and no component depends on unproven or experimental infrastructure.

5.2 Operational Feasibility

Operational feasibility assesses whether the platform fits naturally into the working patterns of its intended users — students and placement coordinators — and whether it can be sustained without disrupting existing processes.

Dimension	Assessment	Residual Risk
User adoption	Low friction — students already hold a college email account and a Google account; sign-in requires no new credentials to create or remember.	Low

Dimension	Assessment	Residual Risk
Administrative workflow fit	Approve / Reject / Edit workflow mirrors how placement coordinators already vet data informally; the system formalizes an existing manual review habit rather than introducing a new one.	Low
Staffing for ongoing operation	Requires at least one Super Admin to manage whitelisting and a small number of whitelisted admins (placement cell staff or senior student volunteers) to review incoming reports.	Medium
Training requirement	Minimal — the interface mirrors familiar dashboard and table patterns; the admin panel's tab structure (Analytics, Reports, Add Company, Add Report, OA Questions, Admins) is self-explanatory to non-technical staff.	Low
Data quality dependency	The system's value depends on timely, honest submission and verification of placement reports each academic year; a lapse in admin review activity would reduce data freshness over time.	Medium
Institutional continuity	As a student-built platform, long-term maintenance depends on knowledge transfer to successive student teams or eventual handover to a permanent TPC-managed resource.	Medium

Overall, the project is judged operationally feasible. The sign-in experience adds no new burden for students, and the admin approval workflow closely follows how placement data was already being informally vetted before the platform existed. The main operational risk is continuity of admin engagement and long-term maintainership, both of which are organizational rather than technical concerns and can be addressed by formally assigning the Super Admin role to a permanent TPC staff position rather than a rotating student volunteer.

6. Conclusion

The problem addressed by the STC Portal — unreliable, fragmented, and unverified placement data with no institution-level access control — is well defined and directly experienced by the student body it serves. The proposed objectives and scope map each identified pain point to a specific platform capability, from domain-restricted authentication to a verified-report approval pipeline. The technical feasibility analysis shows that every architectural component relies on mature, well-supported technology, and the operational feasibility analysis shows that the platform's workflows extend rather than disrupt the placement cell's existing practices. On this basis, the STC Portal is assessed as both technically and operationally feasible for continued development and long-term adoption at KIET Group of Institutions.